Krishnaa **Vadivel** | Software Engineer | Machine Learning | Distributed Computing

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Summary

Software engineer and Yale PhD researcher building machine-learning, distributed-computing, and automation workflows in Python and C++. Experience spans CUDA/ROCm acceleration, national-lab HPC systems, numerical software, Linux-based tooling, and agentic development workflows using modern LLM APIs and local models.

Experience

Yale University, Electrical Engineering
PhD Research Assistant — *2021–present*

- Develop Python/C++ workflows for first-principles and many-body simulation, using MPI, OpenMP, PETSc, SLEPc, CUDA, ROCm, CMake, and Linux shell tooling across leadership-class HPC environments.
- Build reusable distributed-computing and machine-learning workflows for materials and molecular problems on national-lab GPU clusters including Frontier, Frontera, and Perlmutter.
- Work across research code, debugging, compiler-aware build workflows, documentation, and theory-to-implementation translation for scalable computational software.
- Co-authored a 2025 *Journal of the American Chemical Society* paper and presented APS talks in 2024, 2025, and 2026, demonstrating research output alongside production-style software work.

Probe Technologies Inc. / Weatherford Wireline Services
Software and Embedded Systems Engineer — *2019–2021*

- Built CUDA-accelerated software for sensor-data and waveform processing, diagnostic analytics, and engineering tooling for large field datasets.
- Developed ASP.NET and Microsoft SQL Server workflows for internal software, full-stack web server management, and remote tool-calibration support.
- Developed C# GUI applications for preparing customer inputs and interacting with engineering workflows tied to deployed hardware systems.

FindLight
Photonics Technical Writer — *2018–2019*

- Wrote technical content translating specialized optics and photonics topics into clear engineering-facing explanations.

Selected Projects

Equivariant Graph Neural Networks: Built PyTorch Geometric workflows for molecular-property prediction with equivariant graph neural networks and graph-transformer-style modeling ideas.

Agentic Coding and Scientific Automation: Built LangChain-based workflows using OpenAI API models and local Ollama models to read documentation, generate structured inputs, and accelerate coding and research workflows.

Distributed Scientific Tooling: Reusable software workflows for simulation setup, engineering analysis, and container-oriented research infrastructure.

GPU Data Systems: CUDA-accelerated Python and C++ utilities for high-throughput data processing, diagnostics, and numerical inspection.

Education

Yale University
PhD, Electrical Engineering — *2021–present*

Yale University
MPhil, Electrical Engineering *2023*

Yale University
MS, Electrical Engineering *2023*

Cornell University
MEng, Electrical and Computer Engineering *2017–2018*

Cornell University
BS, Electrical and Computer Engineering *2013–2017*

Technical Skills

Languages: Python, C, C++, JavaScript, Bash, SQL

Scientific Computing: MPI, OpenMP, PETSc, SLEPc, NumPy, pandas, SciPy, SymPy, matplotlib

Systems: Linux, CMake, Docker, Docker Compose, Kubernetes, gcloud

Platforms: CUDA, ROCm, Frontier, Frontera, Perlmutter

ML / Automation: PyTorch, PyTorch Geometric, scikit-learn, LangChain, OpenAI API, Ollama, graph neural networks, graph transformers, agentic workflows